Year 11 Foundation

REVISION PACK

Handling Data 4

Name: _____

Class: _____

Handling Data 4 · Collecting and presenting data (Lessons 4.1, 4.2, 4.3, 4.4)

Worked example: frequency tables

Here are the numbers of cars owned by 10 families. 2 0 1 1 3 1 2 1 0 2

(a) Complete a frequency table. (b) Write down the mode. (c) Work out the range. (d) Work out the mean.

(a)	**Step 1**	Tally and count	$0:2,\ 1:4,\ 2:3,\ 3:1$
(b)	**Step 1**	Most common	1
(c)	**Step 1**	Biggest − smallest	$3-0=3$
(d)	**Step 1**	Value × frequency	$0+4+6+3=13$
	Step 2	Divide by 10	$13\div 10=1.3$

Practice: frequency tables

1.1

Number	Tally	Frequency
0		
1		
2		
3		
4		

Here are the numbers of brothers and sisters of 20 students.

1 2 0 1 3 2 1 0 2 1

1 4 2 1 0 3 1 2 1 2

(a) Complete the frequency table.

(b) Write down the mode.

(c) Work out the range.

(d) Work out the mean number of brothers and sisters.

1.2

Number of books	Tally	Frequency
0		
1		
2		
3		
4		

Here are the numbers of books read last month of 20 students.

2 3 1 0 2 2 4 1 3 2

0 1 2 3 2 1 2 4 3 2

(a) Complete the frequency table.

(b) Write down the mode.

(c) Work out the range.

(d) Work out the mean number of books.

1.3

Number of goals	Tally	Frequency
0		
1		
2		
3		
4		

Here are the numbers of goals scored of 20 students.

0 1 1 2 0 3 1 1 0 2

1 4 1 0 2 1 3 1 2 1

(a) Complete the frequency table.

(b) Write down the mode.

(c) Work out the range.

(d) Work out the mean number of goals.

Practice: frequency tables (continued)

1.4

Number of cousins	Tally	Frequency
0		
1		
2		
3		
4		

Here are the numbers of cousins of 20 students.

2 1 3 0 2 4 1 2 3 2

1 0 2 3 1 2 4 2 1 3

(a) Complete the frequency table.

(b) Write down the mode.

(c) Work out the range.

(d) Work out the mean number of cousins.

1.5

Number of drinks	Tally	Frequency
0		
1		
2		
3		
4		

Here are the numbers of drinks they had yesterday of 20 students.

1 0 2 1 1 3 2 1 0 1

2 1 4 1 0 2 1 3 1 2

(a) Complete the frequency table.

(b) Write down the mode.

(c) Work out the range.

(d) Work out the mean number of drinks.

1.6

Number of computers	Tally	Frequency
0		
1		
2		
3		
4		

Here are the numbers of computers at home of 20 students.

0 2 1 1 0 3 1 2 0 1

1 0 2 1 4 1 0 2 1 1

(a) Complete the frequency table.

(b) Write down the mode.

(c) Work out the range.

(d) Work out the mean number of computers.

Handling Data 4 · Collecting and presenting data (Lessons 4.1, 4.2, 4.3, 4.4)

Worked example: pie charts from a table

Information about 24 students is shown in a pie chart. 6 of the 24 students walk to school. Work out the angle of the sector for walking.

Step 1	One student	$360 \div 24 = 15^\circ$
Step 2	Multiply	$6 \times 15 = 90^\circ$

Practice: pie charts from a table

1.7 Information about 30 students is shown in a pie chart. 7 of the 30 students cycle to school. Work out the angle of the sector for cycling.

1.8 Information about 24 students is shown in a pie chart. 10 of the 24 students have one brother. Work out the angle of the sector for one brother.

1.9 Information about 40 students is shown in a pie chart. 14 of the 40 students have two pets. Work out the angle of the sector for two pets.

1.10 Information about 18 students is shown in a pie chart. 5 of the 18 students play tennis. Work out the angle of the sector for tennis.

1.11 Information about 45 students is shown in a pie chart. 12 of the 45 students read every day. Work out the angle of the sector for reading every day.

1.12 Information about 60 students is shown in a pie chart. 22 of the 60 students take the bus. Work out the angle of the sector for the bus.

Handling Data 4 · Collecting and presenting data (Lessons 4.1, 4.2, 4.3, 4.4)

Worked example: the median of six numbers

Find the median of these numbers. 4 9 1 7 6 3

Step 1	In order	$1, 3, 4, 6, 7, 9$
Step 2	Two middle numbers	4 and 6
Step 3	Halfway	$(4 + 6) \div 2 = 5$

Practice: the median of six numbers

1.13 Find the median of these numbers. 5 2 8 4 10 7

1.14 Find the median of these numbers. 12 3 9 6 15 4

1.15 Find the median of these numbers. 7 1 5 9 3 6

1.16 Find the median of these numbers. 10 14 8 12 6 11

1.17 Find the median of these numbers. 2 9 4 9 5 1

1.18 Find the median of these numbers. 20 15 18 11 16 13

Handling Data 4 · Collecting and presenting data (Lessons 4.1, 4.2, 4.3, 4.4)

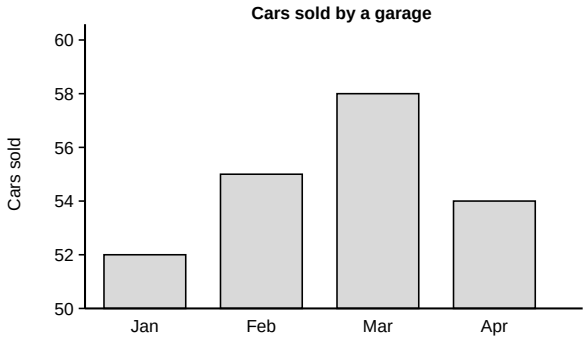
Worked example: misleading charts

A bar chart shows the numbers of cars sold. Its vertical axis starts at 50. Give one reason why the bar chart is misleading.

Step 1	Look at the axis	it starts at 50, not 0
Step 2	Answer	small differences look big

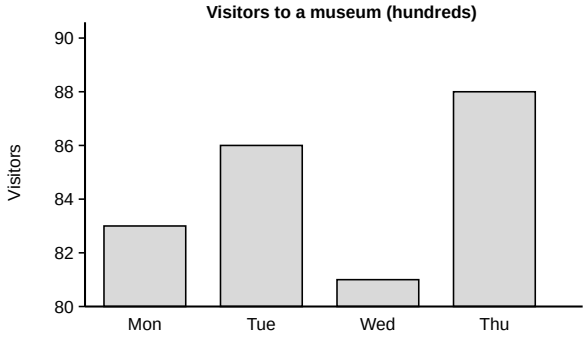
Practice: misleading charts

1.19



The bar chart shows cars sold by a garage.
Give one reason why the bar chart is misleading.

1.20



The bar chart shows visitors to a museum (hundreds).
Give one reason why the bar chart is misleading.

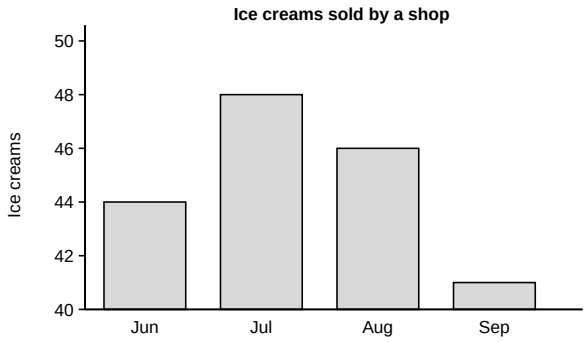
1.21



The bar chart shows pizzas sold by a shop.
Give one reason why the bar chart is misleading.

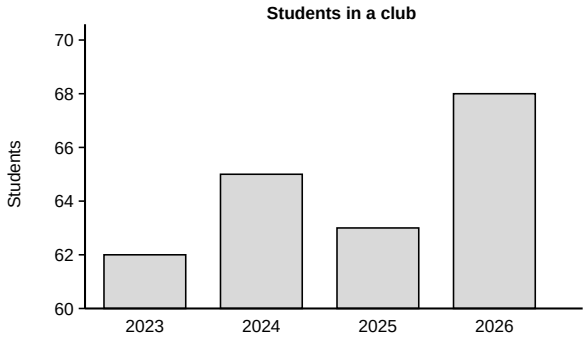
Practice: misleading charts (continued)

1.22



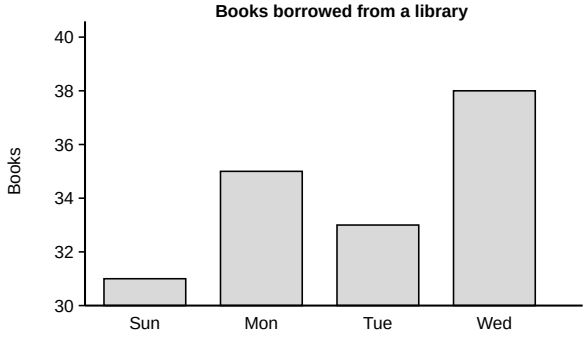
The bar chart shows ice creams sold by a shop.
Give one reason why the bar chart is misleading.

1.23



The bar chart shows students in a club.
Give one reason why the bar chart is misleading.

1.24



The bar chart shows books borrowed from a library.
Give one reason why the bar chart is misleading.

Handling Data 4 · Collecting and presenting data (Lessons 4.1, 4.2, 4.3, 4.4)

Worked example: questionnaires

Ali uses this question on a questionnaire to find out how much students spend on snacks. Write down two things that are wrong with the question.

How much do you spend on snacks?
0–1 KD ☐ 1–2 KD ☐ 2–3 KD ☐

Step 1	Is there a time?	No: per day or per week?
Step 2	Do boxes overlap?	Yes: 1 KD is in two boxes
Step 3	Is every answer there?	No box for more than 3 KD

Practice: questionnaires

1.25 Huda uses this question on a questionnaire to find out how much time students spend on homework. Write down two things that are wrong with the question.

How many hours of homework do you do?
0–2 ☐ 2–4 ☐ 4–6 ☐

1.26 Majed uses this question on a questionnaire to find out how often people exercise. Write down two things that are wrong with the question.

How many times do you exercise?
1–3 ☐ 3–5 ☐ 5–7 ☐

1.27 Layla uses this question on a questionnaire to find out how much people spend on lunch. Write down two things that are wrong with the question.

How much do you spend on lunch?
0–2 KD ☐ 2–4 KD ☐ 4–6 KD ☐

Practice: questionnaires (continued)

1.28 Omar uses this question on a questionnaire to find out how much time people spend on their phones. Write down two things that are wrong with the question.

How many hours do you spend on your phone?
0–2 ☐ 2–4 ☐ 4–6 ☐

1.29 Mona uses this question on a questionnaire to find out how much people spend on clothes. Write down two things that are wrong with the question.

How much do you spend on clothes?
0–10 KD ☐ 10–20 KD ☐ 20–30 KD ☐

1.30 Fahad uses this question on a questionnaire to find out how many books students read. Write down two things that are wrong with the question.

How many books do you read?
1–2 ☐ 2–3 ☐ 3–4 ☐

Handling Data 4 · Collecting and presenting data (Lessons 4.1, 4.2, 4.3, 4.4)

Worked example: pie charts

40 people were asked to name their favourite fruit. 12 chose apples. The results are shown in a pie chart. Work out the angle of the sector for apples.

Step 1	One person	$360 \div 40 = 9^\circ$
Step 2	Multiply	$12 \times 9 = 108^\circ$

Practice: pie charts

1.31 30 people were asked to name their favourite colour. 11 chose blue. The results are shown in a pie chart. Work out the angle of the sector for blue.

1.32 45 people were asked to name their favourite drink. 16 chose juice. The results are shown in a pie chart. Work out the angle of the sector for juice.

1.33 60 people were asked to name their favourite subject. 25 chose maths. The results are shown in a pie chart. Work out the angle of the sector for maths.

1.34 40 people were asked to name their favourite fruit. 9 chose bananas. The results are shown in a pie chart. Work out the angle of the sector for bananas.

1.35 72 people were asked to name their favourite sport. 20 chose tennis. The results are shown in a pie chart. Work out the angle of the sector for tennis.

1.36 20 people were asked to name their favourite pet. 7 chose cats. The results are shown in a pie chart. Work out the angle of the sector for cats.

Handling Data 4 · Averages and range (Lesson 4.5)

Worked example: mean and range

Here are five numbers. 5 9 4 8 4

(a) Work out the mean. (b) Work out the range.

(a)	**Step 1**	Add	$5 + 9 + 4 + 8 + 4 = 30$
	Step 2	Divide by 5	$30 \div 5 = 6$
(b)	**Step 1**	Biggest − smallest	$9 - 4 = 5$

Practice: mean and range

2.1 Here are five numbers. 7 3 9 5 6

(a) Work out the mean. (b) Work out the range.

2.2 Here are five numbers. 12 8 15 10 5

(a) Work out the mean. (b) Work out the range.

2.3 Here are five numbers. 2 9 4 7 3

(a) Work out the mean. (b) Work out the range.

2.4 Here are five numbers. 6 11 8 10 5

(a) Work out the mean. (b) Work out the range.

2.5 Here are five numbers. 14 9 12 7 8

(a) Work out the mean. (b) Work out the range.

2.6 Here are five numbers. 3 10 5 8 4

(a) Work out the mean. (b) Work out the range.

Handling Data 4 · Averages and range (Lesson 4.5)

Worked example: mode and median

Here are seven numbers. 4 6 6 2 9 6 7

(a) Write down the mode. (b) Find the median.

(a)	**Step 1**	Most common	6
(b)	**Step 1**	In order	$2, 4, 6, 6, 6, 7, 9$
	Step 2	Middle number	6

Practice: mode and median

2.7 Here are seven numbers. 4 6 4 8 9 4 5

(a) Write down the mode. (b) Find the median.

2.8 Here are seven numbers. 7 3 3 5 9 3 6

(a) Write down the mode. (b) Find the median.

2.9 Here are seven numbers. 2 8 5 8 1 8 4

(a) Write down the mode. (b) Find the median.

2.10 Here are seven numbers. 10 7 12 7 9 11 7

(a) Write down the mode. (b) Find the median.

2.11 Here are seven numbers. 6 2 9 2 5 3 2

(a) Write down the mode. (b) Find the median.

2.12 Here are seven numbers. 15 12 15 18 11 15 14

(a) Write down the mode. (b) Find the median.